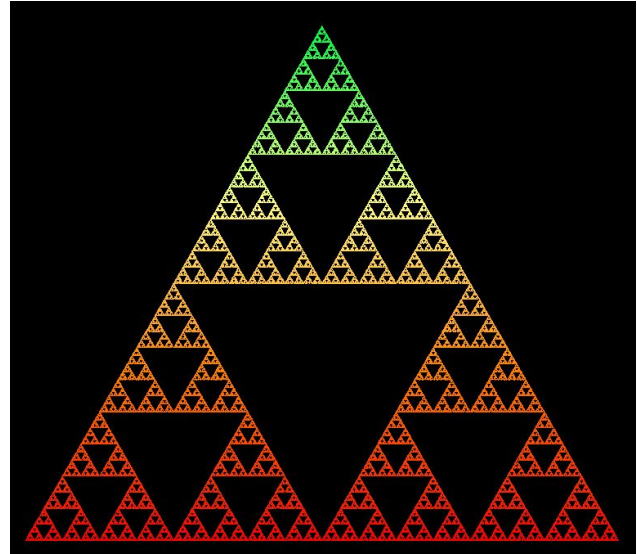


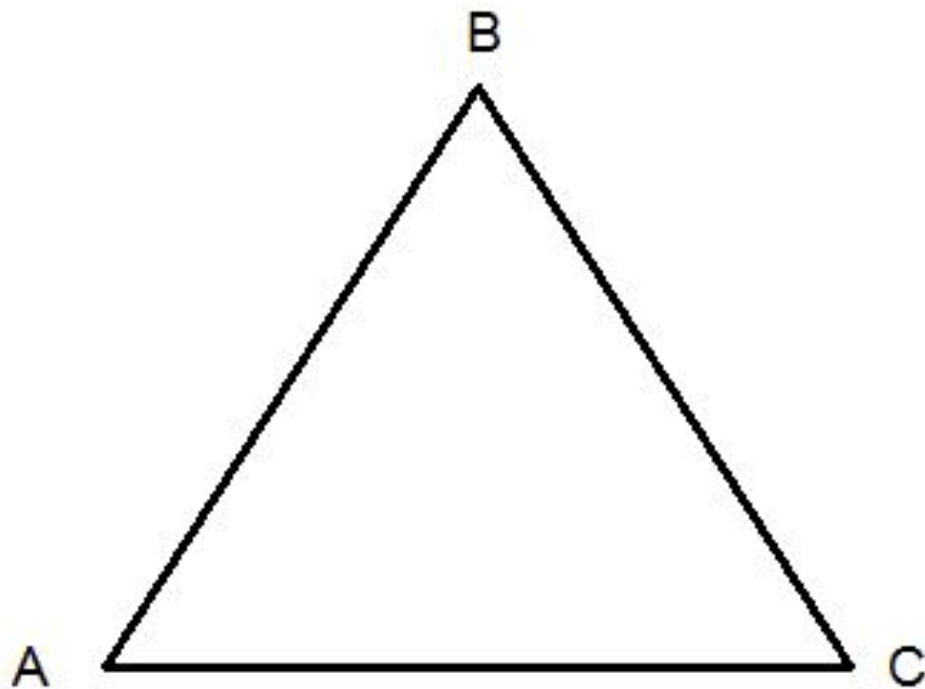
Chaos Game And Serpinski triangle



memes.com - memes.com



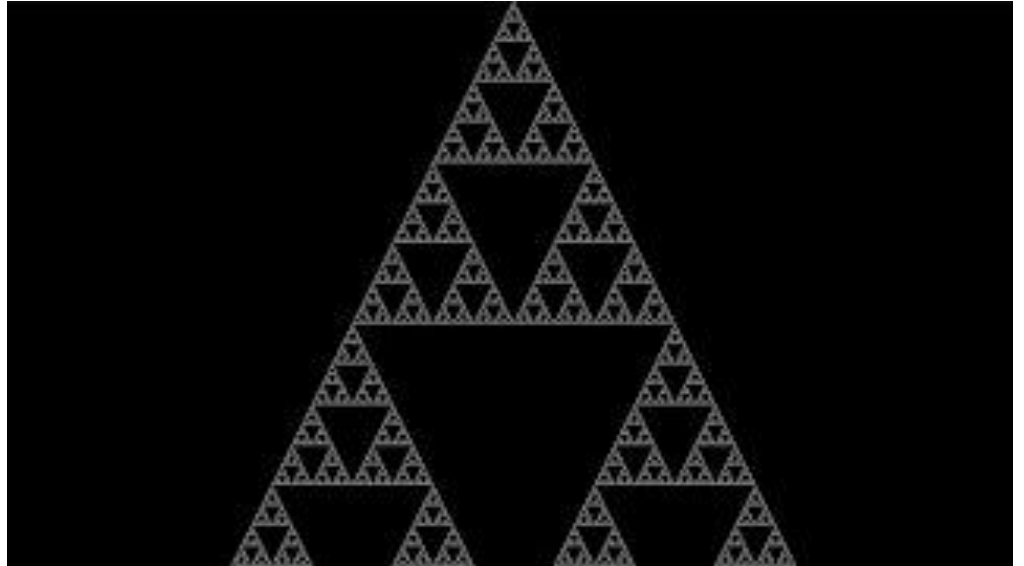
Chaos Game



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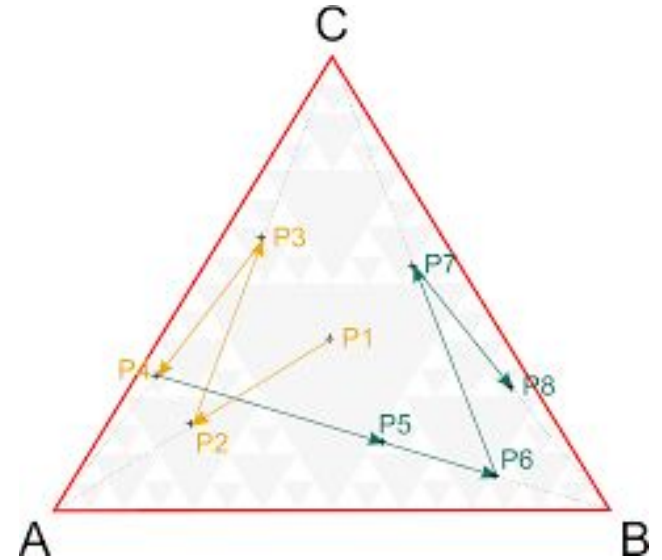
Sierpinski triangle

Named after Polish mathematician **Wacław Sierpiński** (1915), but the pattern appears in art and architecture centuries earlier.



Why Sierpinski triangle from chaos game?

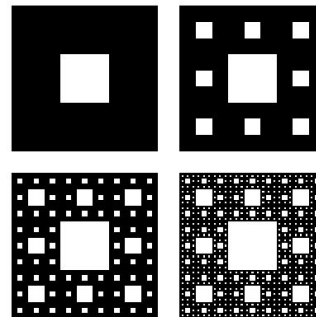
Induction



Some more chaos games

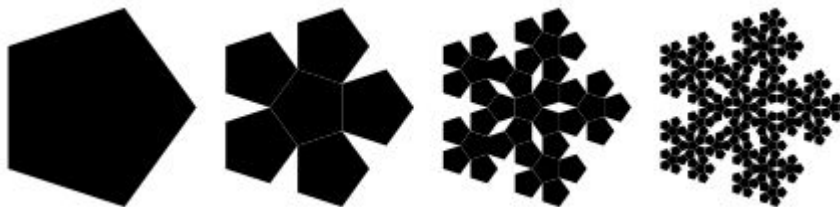
Square $\rightarrow$ Sierpiński Carpet

- 4 vertices of a square, move $\frac{2}{3}$ of the way toward chosen vertex
- Produces a square with the middle third removed, recursively



Pentagon $\rightarrow$ Pentaflake

- 5 vertices of a pentagon, move $\frac{1}{2}$ way toward chosen vertex
- BUT: skip the vertex if it was chosen last time (restriction rule)



Weed



More serpinski sneak peak

<https://www.mathsisfun.com/games/towerofhanoi.html>



Binomial Theorem

$$(a + b)^0 = 1$$

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

